11-Façade

- The **Facade Pattern** is a **structural design pattern** used in software engineering. It provides a **simplified interface** to a complex system of classes, libraries, or frameworks.

- ដើម្បីផ្តល់ភាពងាយស្រួលក្នុងការគ្រប់គ្រងដោយមិនចាំបាច់ធ្វើការគ្រប់គ្រងទៅលើ Class ដទៃទៀត ដោយយើងធ្វើការគ្រប់គ្រងតែទៅលើ interface

🔍 What problem does it solve?

When a system becomes complex, interacting with it directly can be confusing and tightly coupled. The Facade pattern:

- Hides complexity
- Provides a clean, easy-to-use interface
- Reduces dependencies between client code and subsystems

💡 Simple Idea

Instead of the client interacting with multiple classes, it interacts with **one single class (the facade)** that internally calls the required components.

🧠 Real-World Analogy

Think of a **home theater system**:

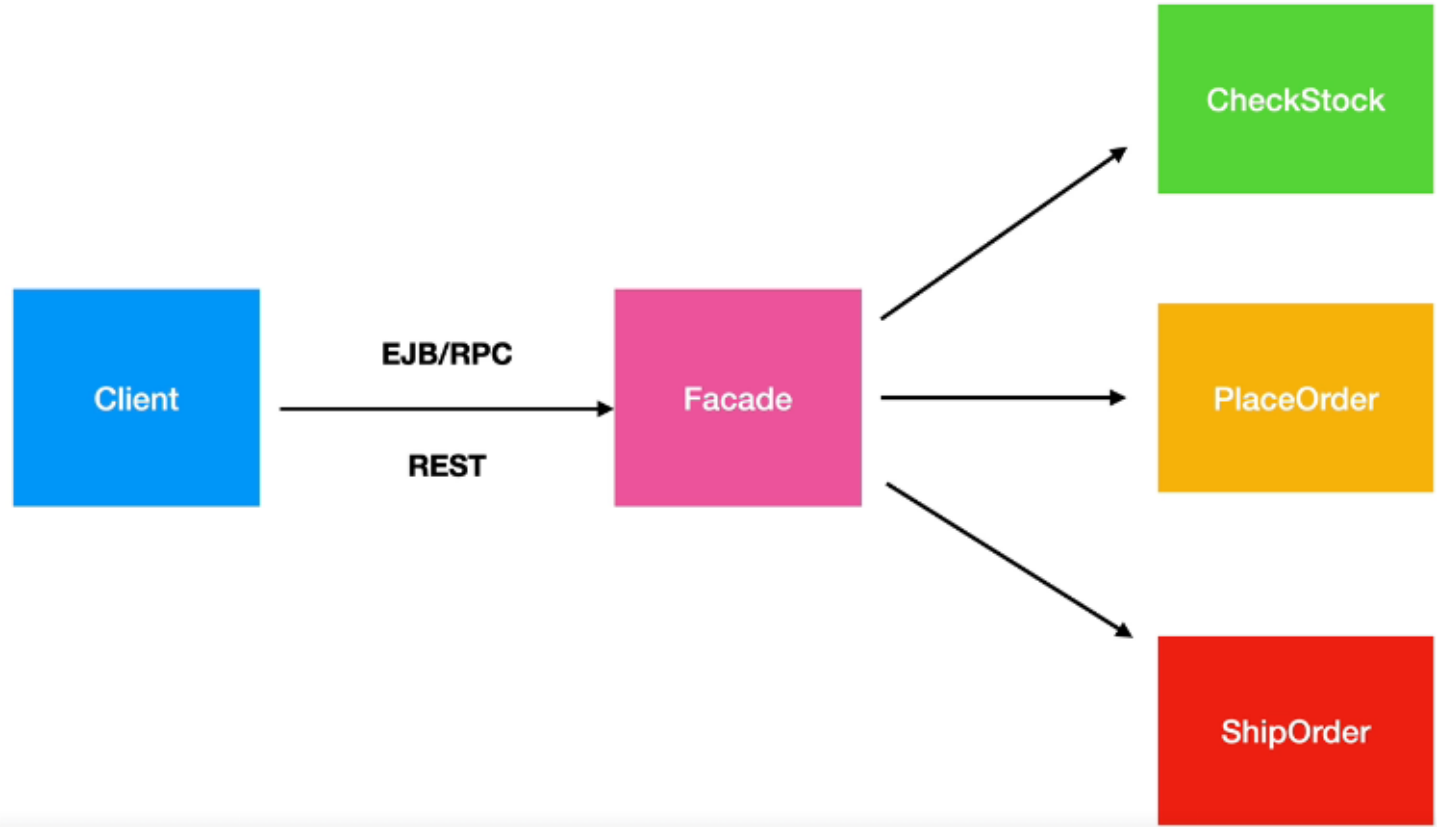
- You have DVD player, speakers, projector, lights, etc.
- To watch a movie, you'd need to turn on everything individually.

👉 A **facade** is like a **remote control with a "Watch Movie" button**:

- It internally coordinates all devices
- You just press one button

📐 Structure

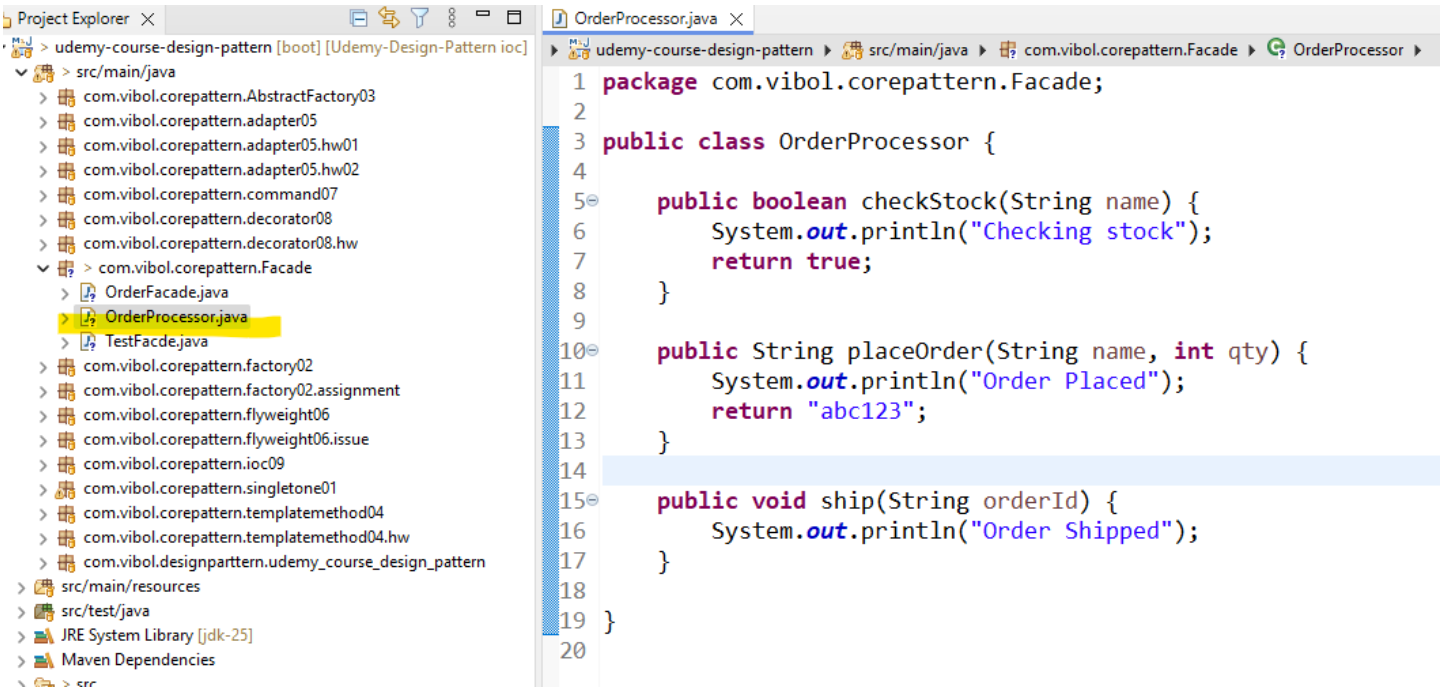
- **Facade** → Simplified interface
- **Subsystem Classes** → Complex underlying components
- **Client** → Uses facade instead of direct interaction



- It hid the complexity of the system by creating layers in the between.

Example:

1-Class



2- Middle class:

OrderProcessor.java | TestFacde.java | OrderFacade.java

udemy-course-design-pattern ▸ src/main/java ▸ com.vibol.corepattern.Facade ▸ OrderFacade ▸

```java
package com.vibol.corepattern.Facade;

public class OrderFacade {

    public void processOrder(String name, int qty) {
        OrderProcessor processor = new OrderProcessor();

        if (processor.checkStock(name)) {
            String orderId = processor.placeOrder(name, qty);
            processor.ship(orderId);
        }
    }
}
```

3-Testing

OrderProcessor.java | TestFacde.java | OrderFacade.java

udemy-course-design-pattern ▸ src/main/java ▸ com.vibol.corepattern.Facade ▸ TestFacde ▸

```java
package com.vibol.corepattern.Facade;

public class TestFacde {

    public static void main(String[] args) {

        OrderFacade facade = new OrderFacade();
        facade.processOrder("MacBook", 3);

    }

}
```

Problems | Javadoc | Declaration | Terminal | Console | Boot Dashboard

<terminated> TestFacde [Java Application] C:\Program Files\Java\jdk-25\bin\javaw.exe (Apr 27, 2026, 9:35:58 AM – 9:35:59 AM) [pid: 17592]

```
Checking stock
Order Placed
Order Shipped
```